Summary

Reference Number

39262

Lead Applicant

Mayur Murali

Research Title

BREATHE: **"B**etter **R**espiratory w**EA**ning using **T**ransparent mac**H**ine learning and data visualisation for invasiv**E** mechanical ventilation"

1. Application Summary Information

Host organisation (which will administer any award)

Imperial College of Science, Technology & Medicine

Application type

NIHR Academy Strategic Theme

Health data science

Partnership Fellowship

N/A

WTE %

1.0

Duration

Doctoral Fellowship is a r-year award and based on your WTE preference the duration in months is calculated as follows.

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Research Title

BREATHE: **"B**etter **R**espiratory w**EA**ning using **T**ransparent mac**H**ine learning and data visualisation for invasiv**E** mechanical ventilation"

Research Type

Pilot study

Secondary data analysis

Qualitative study

Proposed start date if grant awarded

1/10/25

2. Applicant CV

Degrees & Professional Qualifications

2024 **FFICM Final MCQ**, Faculty of Intensive Care Medicine
2024 **MSc: Health Data Science**, University College London
Merit awarded
2021 **FRCA**, Royal College of Anaesthetists
2019 **Fellowship of the Higher Education Academy**, Higher Education Academy
2019 **MRCP (UK)**, Royal College of Physicians
2018 **MSc: Masters in Medical Education**, University College London/Royal College of Physicians
Merit awarded
2016 **Diploma in Tropical Medicine & Hygiene**, Liverpool School of Tropical Medicine
Merit awarded
2013 **Primary Medical Qualification (MBBS): Barts & The London School of Medicine & Dentistry**
Merit in Clinical Specialties
2008 **BSc: University of Warwick**
Bachelor of Science (Hons) in Biological Sciences; Class 2:1

Present & previous positions

Research grants held

Nil

Publications record

2024 "**Wellbeing: paved with good intentions, but the road needs fixing first"** – **Murali M**, Agarwal S
Anaesthesia (published online)
Pubmed ID: 39415751
doi: 10.1111/anae.16410

2024 "**Large language models and artificial intelligence: the coming storm for academia"** – **Murali M**, Wiles M.
Anaesthesia (published online)
Pubmed ID: 39316447
doi: 10.1111/anae.16441

2024 "**Long COVID symptoms and demographic associations: A retrospective case series study using healthcare application data"**
Sunkersing D, Goodfellow H, Mu Y, Ramasawmy M, **Murali M** et al.
Journal of the Royal Society of Medicine Open **15** (7) 1-12
PubMed ID: 39228407
doi: 10.1177/20542704241274292

2024 **"Clinical practice, decision-making, and use of clinical decision support systems in invasive mechanical ventilation: a narrative review"** – **Murali M**, Ni M, Karbing DS, Rees SE, Komorowski M, Marshall D, Ramnarayan P, Patel BV.

BJA **133** (1): 164-177
Pubmed ID: 38637268
doi: 10.1016/j.bja.2024.03.011

2024 **"How to write pilot and feasibility studies"** – **Murali M,** Charlesworth M.
Anaesthesia **79** (6): 650-655
Pubmed ID: 38489827
doi: 10.1111/anae.16283

2023 **"Early warning scores: the case for aggregate vs single extreme parameter activation to detect patient deterioration"** – **Murali M**, Inada-Kim M.
Anaesthesia **78**: 803-806
Pubmed ID: 37195103
doi: 10.1111/anae.16048

2023 "**Ethnic minority representation in UK COVID-19 trials: systematic review & meta-analysis" – Murali M**, Gumber L, Jethwa H, Ganesh D, Hartmann-Boyce J, Sood H, Zaccardi F, Khunti K.
BMC Medicine **21**: 111
Pubmed ID: 36978166
doi: 10.1186/s12916-023-02809-7

2022 "**Frameworks for implementation, uptake and use of digital healthcare interventions in ethnic minority populations: a scoping review using cardiometabolic disease as a case study"** – Ramasamy M, Poole L, Thorlu-Bangura Z, Chauhan A, **Murali M** et al.
JMIR Cardio **6** (2): e37360.
PubMed ID: 35969455
doi: 10.2196/37360

2020 **"Diarrhoea in critical care is rarely infective in origin, associated with increased length of stay and higher mortality"** – **Murali M**, Ly C, Tirlapur N, Montgomery HE, Cooper JA, Wilson AP.
Journal of the Intensive Care Society **21**(1): 72-78.
PubMed ID: 32284721
doi: 10.1177/1751143719843423

2021 **"Medical leadership in the military and NHS: what transferrable lessons can be identified?"** – Davies R, **Murali M**.
BMJ Leader **5**: 232-235.
doi: 10.1136/leader-2020-000439

2019 **"A service evaluation of specialist nurse follow-up of bowel cancer patients after surgery"** – Mole G, **Murali M**, Carter S, Gore D, Broadhurst J, Moore T, Vickers P, Miles A.
British Journal of Nursing **28** (19): 1134-1138.
PubMed ID: 31647733
doi: 10.12968/bjon.2019.28.19.1234

2017 **"Tackling global medical education inequality through e-learning" –** Murali M.

Medical Education **51**: 556-7 (2017).
PubMed ID: 28394069
doi: 10.1111/medu.13301

Conference Abstracts:

2022 **"Modelling the relationship between skin blood flow, cutaneous heat flux and skin temperature during passive heat stress"** – Mullington CJ, Strutton PH, Hegde SP, Rajadurai A & Murali M.
Proc Physiol Soc **49**: PC42.

2022 **"Concurrent Covid-19 infection amongst the trauma population presenting to a major trauma centre"** – Partington T, Same M, Murali M, Scurr C, Parry J & Illingworth J.
Anaesthesia **77**(2): 30.
doi: 10.1111/anae.15631

2021 **"Right lingual artery bleed due to seizure-induced tongue biting causing acute airway compromise: clinical, technical and human factors considerations"** – Murali M, Scurr C, Kearsley R, Kynoch M
Anaesthesia **76** S5): 85.
doi: 10.1111/anae.15521

2020 "**ZamSim: embedding a structured critical incident simulation programme for physician anaesthetists in Zambia"** – Murali M, Harrington J, Wang TY, Hall D, Mumphansha H, Zyambo M.
EJA **37** (e-Suppl 58): 408-409.

2020 **"Inspiring the next generation of anaesthetists in Zambia through a novel, structured clinical skills programme for medical undergraduates"** – Murali M, Harrington J, Wang TY, Boyd M, Davies R, Hall D.
EJA **37** (e-Suppl 58): 403.

2020 **"Establishing sustainable and regular focused critical care echocardiography training in a developing world Intensive Care Unit"** – Harrington J, Hall D, Wang TY, Murali M, Mwansa S, Hung J.
EJA **37** (e-Suppl 58): 284.

2020 **"A quality improvement project to prevent hyperoxia in a developing world intensive care unit"** – Harrington J, Hall D, Wang TY, Murali M, Mwansa S
Critical Care **24** (Suppl 1): 107.
doi: 10.1186/s13054-020-2772-3

2020 **"Establishing a focused critical care echocardiography programme in a developing world intensive care unit"** – Hall D, Harrington J, Murali M, Wang TY, Shamambo N.
Critical Care **24** (Suppl 1): 201.
doi: 10.1186/s13054-020-2772-3

2018 **"Silver Trauma Pathway: a new gold standard in elderly trauma care"** – Murali M, Bolton L, Bhaktal S.
BMJ Leader **2**: A32.

2018 **"Improving the patient pathway: an audit of elderly trauma management in a busy London teaching hospital"** – Murali M, Bolton L, Bhaktal S.
Anaesthesia **73**: 83. Special Issue – abstracts of the AAGBI GAT Annual Scientific Meeting 2018.
doi: 10.1111/anae.14377

2017 **"Pre-operative testing for elective surgery at busy London teaching hospitals" –** Murali M, Patch J, Chapman M & Koutra M.
Anaesthesia **72**: 70. Special Issue – abstracts of the AAGBI GAT Annual Scientific Meeting 2017.
doi: 10.1111/anae.13974

Study Collaborator:

2020 **"Protocol for direct reporting of awareness in maternity patients (DREAMY): a prospective, multicenter cohort study of accidental awareness during general anaesthesia"** – Odor PM, Bampoe S, Lucas DN, Moonesinghe SR, Andrade J, Pandit JJ.
Int J Obstet Anaes **42**: 47-56.
PubMed ID: 32139144
doi: 10.1016/j.ijoa.2020.02.004

2018 **"Cancelled operations: a 7-day cohort study of planned adult inpatient surgery in 245 UK National Health Service hospitals"** – Wong DJN, Harris SK, Moonesinghe SR.
Br J Anaesthesia **121**(4): 730-738.
PubMed ID: 30236235
doi: 10.1016/j.bja.2018.07.002

Books:

2019 **"Surviving Prescribing"** – Murali M, Shulman R, Montgomery H.
Cambridge University Press.
Editor and chapter author

Study Collaborator:

2024 **"Nitrous Oxide Manifold and Other Reduction of Emissions (NoMoreGas): a multicentre observational study evaluating pre-utilisation loss of nitrous oxide"** – Thomas MAF, Ward CJ, Sinnott ME, Davies TW, Wong JM, Wong JKL, Kunst G, Anwar S.
BJA (published online)
Pubmed ID: 39389833
doi: 10.1016/j.bja.2024.08.027

2024 **"Critical care unit bed availability and postoperative outcomes: a multinational cohort study"** – Campbell RAS, Thevathasan T, Wong DJN, Wilson AM, Lindsay HA, Campbell C, Popham S, Barneto LM, Myles PS, Moonesinghe SR, Harris SK.
Anaesthesia (published online)
Pubmed ID: 39326458

doi: 10.1111/anae.16383

2021 **"Incidence of accidental awareness during general anaesthesia in obstetrics: a multicentre, prospective cohort study"** - Odor PM, Bampoe S, Lucas DN, Moonesinghe SR, Andrade J, Pandit JJ.
Anaesthesia **76**(6): 759-776.
Pubmed ID: 33434945
doi: 10.1111/anae.15385

2021 **"General anaesthetic and airway management practice for obstetric surgery in England: a prospective, multicentre observational study"** – Odor PM, Bampoe S, Moonesinghe SR, Andrade J, Pandit JJ, Lucas DN.
Anaesthesia **76**(4): 460-471.
Pubmed ID: 32959372
doi: 10.1111/anae.15250

2020 **"Protocol for direct reporting of awareness in maternity patients (DREAMY): a prospective, multicenter cohort study of accidental awareness during general anaesthesia"** – Odor PM, Bampoe S, Lucas DN, Moonesinghe SR, Andrade J, Pandit JJ.
Int J Obstet Anaes **42**: 47-56.
PubMed ID: 32139144
doi: 10.1016/j.ijoa.2020.02.004

2018 **"Cancelled operations: a 7-day cohort study of planned adult inpatient surgery in 245 UK National Health Service hospitals"** – Wong DJN, Harris SK, Moonesinghe SR.
Br J Anaesthesia **121** (4): 730-738.
PubMed ID: 30236235
doi: 10.1016/j.bja.2018.07.002

Relevant prizes, awards and other academic distinctions

Awards

2020 First Prize – Imperial School of Anaesthesia Trainee Presentation Day, Imperial School of Anaesthesia
2019 Second Prize – Anaesthesia Section Trainee Prize, Royal Society of Medicine
2018 First Prize for Innovation, Pan London Trauma Symposium
2014 Teaching Excellence Award (Trainee Merit Awards in Foundation), North East Thames Foundation School
2014 Judi Sharpe Scholarship for Teaching and Educational Excellence, Southend Hospital
2013 Teaching Excellence Award 2013, Newham University Hospital
2012 Merit in Clinical Specialties in Fourth Year MBBS Clinical Examinations, Barts & The London Medical School
2012 First Place – Michael Frowen Memorial Essay Prize for Medical Students, Medical Council on Alcohol

Bursaries

2019 IRC Travel Grant, Association for Anaesthetists of Great Britain & Ireland
2019 Society for Education in Anaesthesia Travel Grant, SEA(UK)
2019 Beit Bursary for Junior Doctors, Beit Trust

2018 Education Fund Award, Whittington Hospital
2013 Enid Linder Foundation Elective Bursary, Barts & The London Medical School

3. Application Research Background

Professional background

Medically qualified

Specialty (if Medically Qualified)

Anaesthesia & Intensive Care Medicine

Do you have a GMC number?

Yes

GMC number

7410240

Please describe your research career to date (1000 words)

My research career has involved developing skills in data science, health services and qualitative research. To date, I have 12 academic publications, 12 conference abstracts, and have co-edited and published a book. The next stage of my academic development is to train to become an expert in these disciplines in my area of academic interest, invasive mechanical ventilation (IMV) in critical care.

Research Interests

During my Academic Clinical Fellowship at Imperial College London, I developed research interests and learned from academic supervisors and collaborators. Background work has included a narrative review assessing ventilation practice and decision-making (published in the *British Journal of Anaesthesia*); a national survey assessing use of clinical decision support systems (CDSS) in intensive

care; and preliminary work building a machine learning model using ventilation data from a randomised controlled trial of over 200 patients with collaborators at Aalborg University. I have developed skills in data science, completing a Masters in Health Data Science at UCL while working full-time clinically, for which I was awarded a Merit. I have learned to code in Python, SQL, Matlab, Stata and R, and gained experience of statistics and machine learning. I am a national coordinator of GEMINI, an international study evaluating ventilation practice, through which I have experienced the Health Research Authority approval process and collaborated with NHS Trusts to contribute to a database of over 10,000 patients, which will act as a data repository for my PhD. The first abstract from GEMINI will be presented at the American Thoracic Society Conference in 2025.

I have developed collaborations with the Imperial-NIHR HealthTech Research Centre, with support provided by methodologists, health economists and implementation scientists; the Imperial Data Science Institute, to assist with machine learning analysis and visualisation development; and Aalborg University, drawing on their expertise in CDSS and biomedical engineering. For GEMINI, I have collaborated with the VentilaGroup in Spain. Additionally, background work has involved collaboration with the Intensive Care Society and the Paediatric Critical Care Society.

I have collaborated with patients through presentations workshops organised by the Research Partner's Group and the Patient Experience Research Centre at Imperial. I have recruited a patient co-applicant and convened a Patient and Family Advisory Group to advise on the project through the charities ICUSteps and Voice. These experiences have refined the proposal, lay summary and outcome measures.

Education & Qualitative Research Experience

I have completed a Masters in Medical Education (UCL), achieving a Merit. Using these skills, I developed my own teaching, winning prizes and being awarded Fellowship of the Higher Education Academy. I developed the Health Professions Education Masters course with Kings College London for healthcare students in Somaliland, teaching virtually and in person; undertook a teaching fellowship in Zambia, for which I was awarded several competitive travel bursaries, delivering simulation and skills workshops, the impact of which have been presented internationally; and co-written online modules for BMJ Learning ("Challenges in Hospital Care: Sepsis – recognition and management in adults") and the Association of Anaesthetists ("Professional Practice in Resource Poor Settings"). I co-edited and wrote several chapters for a textbook, "Surviving Prescribing", published by Cambridge University Press. I am on the committee of two courses delivered through the Imperial School of Anaesthesia, the Novice Course, and the Imperial Final Viva Course. Through the Masters, I developed critical appraisal and qualitative analysis skills, and learned how to create data collection tools, conduct interviews, and lead focus groups.

Leadership

I have undertaken a Trainee Editor Fellowship for the journal *Anaesthesia*, reviewing over 40 manuscripts, contributing to the Editorial Board and presenting at conferences. I have experience as a peer reviewer for the journals *Medical Education, BMJ Open Respiratory Research* and the *Journal of the Intensive Care Society*. I have collected data for national perioperative studies (SNAP-2, DREAMY and PQIP), and am listed as a study collaborator on 6 published manuscripts. I hold several leadership positions, including Chair of the Pan-London Perioperative Audit and Research Network, organising the Annual

Scientific Meeting, selecting the projects supported by the network, providing research support, and delivering the annual Academic Anaesthesia Regional Training Day. I am the Equality, Diversity and Inclusion (EDI) Representative for the Faculty of Intensive Care Medicine resident doctor committee, setting up an EDI strategy and working on projects to tackle differential attainment. I am a Fellow of the South Asian Health Foundation, undertaking projects analysing health inequalities, which have been published in peer-reviewed journals and developing my understanding of the importance of EDI in research. Previously, I was the Academic Trainee Representative for the Imperial School of Anaesthesia, providing a point of contact for academic trainees and organising the North London Junior Anaesthetic Group conference.

Achievements

I have shown academic aptitude throughout my career, receiving a distinction for my dissertation and an Upper Second-Class Honours overall in my undergraduate degree, a Merit in Clinical Specialties and first place in a national prize while at medical school. During clinical training, I have completed the MRCP, a Diploma in Tropical Medicine & Hygiene, and the FRCA; been awarded several competitive travel bursaries to support an anaesthesia fellowship in Zambia; and received prizes for quality improvement projects. I have completed a National Medical Director's Clinical Leadership Fellowship at NHS England, observing the interaction between policymakers and academics. I contributed to policy documents (e.g., Evidence Based Interventions Programme), worked on national projects and collaborated with professionals from across disciplines. I developed interests in analysing health services and understanding how to navigate complex regulatory and commissioning processes.

Impact of Fellowship

While still at an early stage of my academic development, I believe that my track record of publications, collaboration, and skill attainment provides a strong foundation for the next step of research training. I am highly motivated to become an expert in the field of IMV, using mixed methods to CDSS using explainable machine learning that deliver personalised ventilation. This prestigious NIHR Doctoral Fellowship will act as a springboard for a career as a Clinician Scientist, leading my own research group and attaining a Chair in anaesthesia and critical care.

Have you already registered for a PhD?

Yes

Has this application, or a similar application, previously been submitted to this or any other funding body?

Yes

Mitigating factors

None

4. Plain English summary of research

600 words

Background

Every year, 70,000 patients need breathing support on intensive care. This is delivered through a machine called a ventilator. Treatment with the ventilator requires understanding of patients and their disease. A clinical decision support system (CDSS) is software that improves decision-making. It matches patient and disease features to a database. Treatment options are then suggested to clinicians. The NHS Long Term Plan has suggested CDSS should support patient management.

Deciding when to reduce ventilator support is important. Leaving patients on a ventilator too long can cause lung injury. However, patients must be strong enough to cope without its support. Previous work has shown differences in clinical practice when reducing ventilator support. There is variable success and no agreement on best practice. This suggests CDSS for ventilator management could be beneficial.

Our national survey showed CDSS are not used for ventilator management. This is because there is limited ready-to-use computer software. The survey also showed clinicians do not trust CDSS. This is because their decisions are rarely explained.

Addressing the problem

This research aims to address these two problems. First, by using computer-based methods to find patterns from routine patient data. Hundreds of variables from thousands of patients will be analysed. These methods are

summarised as machine learning (ML). Second, new visual methods will be used to illustrate the data. This will improve understanding, trust and use of CDSS.

Research plan

This research will be conducted in three steps. First, understanding the patient. We will model a patient's disease using ML. This will improve understanding of when to reduce ventilator support. Predictions will be created for the patient's whole admission. The most important features for each prediction will be shown. Models will be checked for performance and bias. We will confirm performance is maintained in under-represented groups.

Second, developing visual methods, or visualisations, to display information. Visualisations will be designed with clinicians who work in intensive care. This will ensure they find the visualisations helpful. The visualisations will illustrate predictions for each patient. They will display the most important features for each prediction. Visualisations will be displayed through an interactive computer system.

Third, assessing clinical impact. Effect on clinical decision-making will be checked through simulation testing. Decision-making using predictions from ML and visualisation will be compared with normal practice. These insights will guide future research. This could include testing CDSS in intensive care.

Patient and public involvement

This project has benefited from continuous patient and public involvement. This has shown this research has potential for significant impact. First, through reducing variation in clinical practice. Second, by improving patient safety. And third, by using visualisations to communicate with patients and families. It has

also shown that this research will benefit the NHS. This is by reducing length of stay and cost of care. It addresses two priority areas from the James Lind Alliance. These have been arrived at through discussion with patients.

Sharing our findings

I will share our findings through local scientific meetings and conferences. Results will be published in academic journals. We will use Imperial College London and collaborator networks to promote our work. This will be visible to industry, other research groups, and stakeholders. I have convened a patient panel. They will help design and present our message to the public. This will be supported by the Imperial College Science Media Team and public outreach activities.

5. Scientific abstract

500 words

Research question

Can explainable machine learning derive trajectories for successful and unsuccessful weaning from invasive mechanical ventilation? Can data visualisation of such explanations aid clinical management?

Background

Invasive mechanical ventilation (IMV) is an important intervention in intensive care, delivered to 70,000 patients annually in the UK. Ventilatory weaning is the process of reducing and removing ventilatory support and shows variable success, with significant heterogeneity in practice. Clinical decision support systems (CDSS) are software that integrate patient data with a computerised knowledge base to provide treatment suggestions. They can support delivery of

personalised ventilation to optimise weaning. However, few CDSS exist, and in feasibility trials, advice adherence is low. A national survey found a key barrier to implementation is that CDSS are “black box”, failing to demonstrate rationale for their decisions.

This project will apply novel machine learning (ML) approaches within large longitudinal clinical datasets to delineate weaning trajectories in adults and children. It aims to test whether explainable AI (XAI) can provide understandable and transparent explanations of decisions and actions to human, through novel data visualisation techniques.

Aims and objectives

1. Iteratively develop and validate XAI-based models to delineate subgroups based on physiological trajectories.
2. Develop and optimise data visualisation approaches to improve model transparency for clinical application.
3. Validate data visualisation approach using simulation.

Methods

I will use a novel ML approach to generate predictions, using three neural network architectures (Temporal Fusion Transformer (TFT); Temporal Convolutional Networks (TCN); Long-Short Term Memory Nets (LSTM)) designed for processing sequential, time-series data to provide multi-horizon forecasts. This architecture will be applied to large longitudinal clinical datasets, totalling over 100,000 patients. An iterative, expert-in-the-loop approach will interrogate the models for bias, validate results and refine forecasts.

Clinician and patient focus groups and co-design workshops will be conducted to understand stakeholder requirements and develop novel data visualisations utilising a web front-end to display the results, displayed through the Data Observatory at the Imperial Data Science Institute. These will be optimised for clinical use through iterative co-design cycles.

Influence on clinical practice and decision confidence will be assessed with a human-AI interaction vignette study using a modified, between-subjects design. Think-aloud techniques will be used to understand rationale for decisions.

Timelines for delivery

I expect to develop and validate XAI-based models in months 1-12; conduct focus and co-design workshops to develop and optimise visualisations in months 6-27; and conduct simulations in months 27-33. I will write up my thesis in the final 6 months of the Fellowship.

Anticipated impact and dissemination

This research will have significant impact. Patients will benefit through receipt of efficacious, evidence-based therapy, clinicians through reduced heterogeneity in practice, and the NHS through optimisation of patient time on the ventilator, reducing length and cost of admission.

Our findings will be disseminated through local scientific meetings, conferences, and publications. I will utilise established collaborator networks of key stakeholders to identify shared learning. I have convened a patient and family advisory group who will assist with co-presenting our message for public dissemination.

6. Detailed research plan

5000 words

1. **What is the problem being addressed?**

   Clinical Need:

   Invasive mechanical ventilation (IMV) is a key supportive therapy for patients on intensive care[1]. In the UK, over 54,000 adults and 13,000 children required IMV in 2021[2,3]. Ventilation decision-making is complex, involving integration of clinical data, setting initial parameters, titrating settings to attain physiological targets, and determining when to reduce ventilator support ("wean"). While several advances have been made in the field[4–6], there is significant heterogeneity in ventilation practice[7]. The goal is to deliver personalised care, tailored to the individual[8].

   Health Technology Solutions:

   Clinical decision support systems (CDSS) for IMV aim to enable personalised care[9]. CDSS match patient characteristics to a computerised knowledge base and generate patient-specific recommendations; they are divided into open-loop (clinician input required to enact suggestion) or closed-loop (operate ventilator independently)[10]. Currently available CDSS for IMV are predominantly based on physiological models[9,11] or rule-based algorithms[12,13].

   Ventilatory weaning is the process of reducing and removing IMV support as patients improve. Our narrative review suggests weaning is where CDSS may be most valuable, as it is multidisciplinary, there is significant heterogeneity, and success rate is low[14]. It is a critical phase that every

ventilated patient must undergo; early liberation reduces the risk of nosocomial pneumonia, ventilator-induced lung injury[15], and prolonged admission[15,16], while premature discontinuation can cause fatigue, impaired gas exchange and reintubation[17,18].

A feasibility study testing an open-loop CDSS from our group showed bedside staff rarely followed CDSS advice[19]. A national survey conducted in response to this implementation gap found clinicians perceive the "black box" aspect of CDSS as particularly troubling, with limited availability of software that can be rapdily implemented in the NHS[20–22]. Moreover, while CDSS for IMV have shown benefit in trials[9,11–13], our survey identified clinical use in the UK is limited.

Enhanced understanding and adoption through data visualisation and explainable AI (XAI):

Machine learning (ML) can be employed to analyse historical patient data and identify patterns that predict outcomes during weaning, though previous studies have not employed time-series data[23–25]. Multi-horizon forecasting (MHF) is the prediction of variables over several points in time. In ventilatory weaning, a single-horizon forecast might predict settings for the next hour, whereas MHF predict parameters and their trajectory over several days or weeks, i.e., at multiple future time steps, providing estimates across the entire admission to enable continuous optimisation of care.

Explainable AI (XAI) is the development of artificial intelligence (AI) models that provide interpretable explanations for their predictions. In the context of MHF, explainability is crucial for understanding the rationale for forecasts. We believe that data visualisation, a technique used to generate hypotheses

from complex data[26], can deliver on the implementation of XAI-based MHF. Visualisation has been used to illustrate patterns in data, communicate findings to expert and lay audiences[27], and explore dynamics of Bitcoin transactions[28], online discussion[29] and bio-surveillance[30]. Data visualisation in intensive care has been limited to live dashboards and handover of care[31–33]. The showcasing of patient ventilatory trajectory to augment decision-making is novel.

Our group has applied XAI to illustrate feature importance when making predictions. Analysis of SHAP (SHapley Additive exPlanations) values derived from time-series models provided insights into which features impacted a model determining patient trajectory in COVID-19 ARDS[7]; simple and static alluvial plots were used to demonstrate how physiological trajectory linked to patient outcome[7].

Hypotheses:

1. XAI-based MHF models can identify weaning trajectories through analysis of longitudinal clinical data.
2. Data visualisation can display these trajectories to augment clinical decision-making.

This project will therefore aim improve understanding of weaning from IMV using ML, creating MHF models that use data visualisation to improve model transparency. This will reduce heterogeneity in weaning practice, augment delivery of personalised ventilatory care, and increase the acceptability of CDSS for clinicians.

2. **Why is this research important in terms of improving the health and/ or the wellbeing of the public and/ or to patients/ service users, carers and health and care services?**

This project tackles an important problem that is recognised by clinicians, policymakers, and patients.

Ventilated patients represent nearly 40% of all adult[2] and 60% of all paediatric admissions to critical care[3] (approximately 70,000 patients[2,3]). The proportion of patients receiving invasive mechanical ventilation in critical care is increasing[34], and invasive mechanical ventilation remains the main reason for patients staying on intensive care[35,36]. Mortality from conditions such as acute respiratory distress syndrome[1] and ventilator associated pneumonia[37] are high, and for patients who are ventilated for long periods, the road to recovery is long and complex[38]. In this context, delivering personalised ventilation strategies, that optimise patient time on the ventilator is becoming increasingly important[8].

Moreover, an increasingly multidisciplinary workforce require tools to assist with delivering safe care[39]. The NHS Long Term Plan has called for the development of decision support tools to help clinicians apply best practice[40], and the Topol Review outlined recommendations to use AI to deliver personalised care[41]. Despite this need, our prior work has shown that CDSS for IMV are rarely used in the UK, and that clinicians and institutions are not ready to use these devices[19].

This is a view shared by our public and patient representatives. I have engaged members of the public through several avenues, including the

charity ICUSteps (https://icusteps.org/), and VOICE (https://www.voice-global.org). I have presented twice to the Research Partner's Group at Imperial, used the Research Support Service, who also have lay representatives, and the Patient Experience Research Centre, both at Imperial, to present to groups with lived experience of both adult and paediatric ventilation. The feedback has been very positive, with members of the public agreeing that this was a very important research topic that could have a substantial impact for patients and their relatives. The feedback has been instrumental in reviewing and amending my lay summary, providing feedback on the proposal, and highlighting outcomes that are important to patients on the ventilator. For example, feedback has highlighted how data visualisations could also be used as a communication aide for conversations with family members and carers, helping to illustrate complex clinical data to improve understanding and assist with achieving treatment concordance. Principle outcomes of importance include reducing time on the ventilator and improving comfort by weaning at the most opportune time, both of which this research will address. They have highlighted the importance of interrogating models for bias, maintaining confidentiality and provided guidance for convening a Patient and Family Advisory Group (PFAG) as part of the study group to ensure that our work remains of relevance to patients, their carers, and members of the public. We will meet every four months to discuss our findings and guide the project moving forward; they will also assist with crafting our message for public dissemination and will be involved in co-presenting our results at conferences. To further assist with this, I have involved a public member co-applicant with experience of participation in previous ventilation studies, who has been instrumental in fine-tuning the lay summary, and will support the project through

participation in co-design workshops, assist with data analysis and presentation of results at conferences.

In addition to these findings, this project also addresses two of the high research priority areas of the James Lind Alliance, which were arrived at through consultation with patients and carers ("What is the best way of ensuring new knowledge and the latest defined standards are introduced into clinical practice in a timely and effective way?" and "How can we predict who will benefit from intensive care before admission and during treatment in the ICU?").

3. **Review of existing evidence – how does the existing literature support this proposal?**

We conducted a narrative review to assess existing evidence, which has been published in the British Journal of Anaesthesia. Our aim was to understand the factors that influence clinical practice and decision-making for mechanical ventilation, both with and without clinical decision support, and the facilitators and barriers to implementation of CDSS. Our database search identified 2,616 studies, of which 30 met the inclusion criteria.

Role allocation was relatively well described, with nurses most likely to lead decisions around weaning readiness and failure; and physicians most likely to lead decisions on initial ventilator settings, selection of weaning method and extubation readiness. Interprofessional collaborative decision-making was considered important, and is similar between adult and paediatric critical care, but more common in western Europe, explained by differences in role allocation, training requirements and cultural factors. Nursing

autonomy and influence on clinical decisions was dependent on nurse: patient ratio and is more likely to be present when that ratio is 1:1. Unit culture and reducing perception of clinical reprimand leads to greater autonomy in clinical decision-making by junior staff[42–51]. However, no information was found on whether this varies with units of different leadership models, size, staff experience and composition.

There was no information on clinical workflow, though one study explored reasons for delay in time to extubate, and no studies defined process measures for ventilation, outside of morbidity, mortality, and weaning success (which itself has several definitions). Arnal and colleagues assessment of INTELLiVENT-ASV, a closed loop CDSS, and showed the potential of these devices to augment workflow: use of CDSS significantly reduced the number manual ventilator setting changes[52].

Weaning of mechanical ventilation was explored in more detail, particularly factors that influence nurse-led weaning. These are broadly grouped into patient, nurse, and system factors[49,53–61]. In contrast, physician-led weaning is poorly understood, with studies describing significant heterogeneity in practice, and no evidence exploring impact of clinician specialty, experience, or level of training. Physicians, while using a combination of subjective and objective information, appeared more likely to draw on “gestalt”[57,62–64]. The success of physician-led weaning is low, and reported to be less than 50% in some studies[14]. Studies using vignettes to assess clinical decision-making have shown significant heterogeneity in the information used to assess patients, clinical practice for the adjustment of ventilator settings, and limited between-clinician agreement[65,66]. The picture is further complicated

by the variability in definitions within critical care and those for treatment criteria[67].

No studies explored decision-making while using CDSS, though several explored facilitators and barriers to implementation of protocols and care bundles in intensive care; these are broadly grouped into those at affecting the clinician, the intervention and the organisation[20–22,68].

One study described factors to support implementation of CDSS; these included integration with existing electronic health records and computer systems, robust monitoring and evaluation, the presence of adequate technical and clinical support and regular interrogation of clinician interaction with the CDSS[20]. Other studies suggested unstructured approaches to the implementation of care bundles[68,69]. No studies documented use of implementation strategies or frameworks to guide transfer to clinical practice.

Our national survey of critical care clinicians has highlighted that CDSS for mechanical ventilation are rarely used in adult and paediatric intensive care units in the UK. While most units have some form of electronic health record, only two CDSS were used in intensive care, neither of which were used often or effectively. CDSS are not used for several reasons, notably because decisions are not always explainable (“black box”), and a lack of business case and evidence-base to support implementation. Despite this, many believe CDSS could be beneficial for improving clinical decision-making, patient safety and outcomes, as well as providing data for audit and research, and acting as a tool to facilitate development of junior staff. Several barriers to implementation were flagged, including resistance from

clinicians to changing practice, a perceived lack of benefit, deskilling of staff and device complexity.

The potential for subgroup identification using clinical and plasma biomarker data has been illustrated by the identification of the hyper- and hypo-inflammatory subgroups in patients with acute respiratory distress syndrome (ARDS)[70], with its potential for therapeutic applications shown in several retrospective analyses[71,72]. Physiological optimisation based on these disease state has also been shown to be effective; patients with diffuse, hyper-inflammatory ARDS are more likely to benefit from higher positive end expiratory pressure (PEEP)[73], reducing dynamic strain and atelectotrauma[74], while those with focal, hypo-inflammatory ARDS benefit from lower PEEP, thus mitigating over-distension of the lung and haemodynamic changes from ventilation[75]. In patients with the most severe ARDS, higher PEEP was associated with better survival[76], while ventilator strategy misaligned to lung morphology was associated with worse survival[77]. In this context, optimising ventilator management through improving our understanding of clinical workflow, mechanical ventilation practice and disease state is key to improving patient management.

Moreover, a study of patients with COVID-19 induced ARDS using longitudinal as well as cross-sectional data identified clinical trajectories in ventilation data that were highly discriminatory for patient prognosis[78]; this approach has been used to demonstrate trajectory of ARDS patients (based on groupings of hypoxaemia resolvers and non-resolvers) from the MIMIC-IV dataset[79]. Further work is required to identify whether longitudinal physiological data can further define groups of ventilated patients.

Our review of the literature and background work in this area has highlighted a lack of understanding of weaning trajectories of patients on mechanical ventilation. While several decision-support systems for mechanical ventilation have been designed[9,11,12,13,80,81], explainability of these models is limited and hinders implementation. The work of our group has shown that critical care units are not yet ready to accept clinical decision support systems for mechanical ventilation[19]. Promising work has been performed that has identified patient trajectories that are predictive of prognosis. We therefore believe that this represents a significant opportunity for research.

4. What is the research question/ aims and objectives?

Research question

Can explainable machine learning derive trajectories for successful and unsuccessful weaning from invasive mechanical ventilation? Can data visualisation of such explanations aid clinical management?

Objectives:

1. Iteratively develop and validate XAI-based models to delineate subgroups based on physiological trajectories.
2. Develop and optimise data visualisation approaches to improve model transparency for clinical application.
3. Validate data visualisation approach using simulation.

5. Project plan

Objective 1:

- To identify subgroups based on patient physiology and response to ventilatory and pharmacological interventions.

- To identify features that define data-driven subgroups.
- To define impact of features with short-, medium- and long-term prediction.

All ventilated patients with sufficient data will be eligible for inclusion in this study. Using a-priori heuristics defining weaning as the first discrepancy between set ventilator and respiratory rate, which have been previously piloted by our laboratory group, I will stratify patients ventilated for < 24 hours, between 1-7, 8-21, and > 21 days. Weaning outcome will be based on heuristics from the literature (success – extubated for > 72h, unassisted with supplementary flow or pressure support > 1 hour after extubation; failure – reintubation or death following extubation).

Patients will be drawn from databases with data sharing agreements in place: MIMIC-IV (40,000 patients), Guy's and St Thomas' (GSTT, 20,000 patients), Royal Brompton & Harefield (RBHT, 20,000 patients) and Imperial College Healthcare NHS Trusts (ICHT, 20,000 patients), and aggregated data from three paediatric intensive care units (PICUs) across ICHT, RBHT and Great Ormond Street NHS Trusts (15,000 patients). Traditional power calculations are not required for machine learning analysis, as by definition there is no specific hypothesis being tested; thus, size of subgroups and the effect size are not known a-priori. MIMIC-IV is hosted on Google Cloud platform as a Structured Query Language (SQL) database; GSTT, RBHT and ICHT data are stored in Trusted Research Environments on secure NHS Servers as SQL databases; and the paediatric ventilation data is stored in the Big Data and Analytics Unit Secure Environment (BDAU SE) as an SQL database. NHS data will be anonymised (MIMIC-IV is already anonymised), and all data will be extracted into CSV files.

The first step will be curation of the datasets. Using clinical domain expertise with support from my supervisors, I will aim to identify a set of core variables that are present across all datasets, with relevant parameters likely to include those in the ventilatory, physiological, and laboratory domains, and patient demographic and admission information. This will ensure that our analyses are performed on clinically universal, rather than site-specific, features. I will undertake pre-processing and exploratory data analysis to understand correlation, collinearity, missingness, and outliers. Each variable in the databases will be interrogated to ensure data is consistent with its label, has appropriate units, and temporal relationships are maintained, and where possible, variables will be standardised and normalised. I will use descriptive statistics and simple visualisations, such as histograms, to improve understanding of complex data. Where significant data missingness is identified, I will use imputation methods, such as parameter-specific time limited forward imputation for time-series data. Noise and outliers will be identified with domain expertise, removing impossible values (e.g., negative physiological values); if necessary, proximity-based (e.g., K-means) or statistical methods (e.g., Z-score) will be used. I will use simple tests of association, such as regression analyses, to identify initial signals in the data.

Initially, each dataset will be analysed individually, to account for variations in populations. I will divide each dataset into training (80%) and test (20%) sets. Using signals identified from regression analyses, I will first assess the performance of simple machine learning architectures, such as Random Forest or support vector machine. I will then assess the Temporal Fusion Transformer (TFT), an attention-based deep neural network architecture for MHF, which provides robust forecast performance, is explainable, and can

be used to derive patterns from time series data. Knowing which factors (feature importance) and historical data points (temporal attention) contribute to predictions will demonstrate rationale. For example, feature importance will be demonstrated using SHAP dependency plots, which illustrate how different features influence model prediction over time; clinicians may therefore be prompted to weigh the significance of parameters depending on point of assessment. For example, in the service evaluation of COVID-19 patients conducted by our group, SHAP dependency plots showed ventilatory ratio was the most influential feature contributing to model performance at day 6[7]. Hence, I will assess whether, and how, ventilatory and pharmacological (e.g., sedatives) interventions affect predictions. I will assess for commonalities in the identification of weaning trajectories and feature importance across datasets; if appropriate, I will combine datasets to create a universal model that can be applied to all adult intensive care populations. However, analysis of paediatric and adult datasets will remain separate due to differences in physiology and ventilation practice between these two populations.

I will promote human-AI collaboration through an iterative, expert-in-the-loop approach. In MHF, domain knowledge is crucial for stakeholders to contribute insights, validate results, and refine forecasting strategies. If the TFT model does not perform as expected, I will combine Temporal Convolutional Networks (TCN) and Long-Short Term Memory Nets (LSTM), convolutional and recurrent neural networks, respectively. The final design will minimise bias while maximising performance and avoiding over-fitting to maintain performance.

Model performance will be assessed in terms of model accuracy (i.e., ability to predict outcomes using a single day's data, data from the whole admission, and across different timeframes of weaning), sensitivity, specificity, F1 score (weighted according to class weights), receiver operating characteristic curve and area under the receiver operating characteristic curve (AUC). I will perform 5-fold cross-validation and select the model with the highest AUC. I will compare different machine learning architectures and select the highest-performing algorithm. Models will be assessed for bias, calibration in under-represented groups, and we will be vigilant for the leak of demographic data (e.g., age, gender) into non-demographic data. I will assess algorithms against international standards for ethical ML and ensure reporting of results conform to SAGER guidelines and NIHR-INCLUDE Ethnicity Framework.

I will collaborate with the Imperial Data Science Institute (DSI) for this analysis.

Objective 2:

- To understand and incorporate key information required for ventilatory weaning and decision-making.
- To understand user requirements for data visualisation and co-design solutions.
- To develop visualisations that display weaning trajectories and features of importance.
- To co-design an effective user interface that enables clinical interpretation of visualisations.

Communication of predictive value, and uncertainty, is critical when conveying forecast reliability and limitations. Uncertainty arises from inherent variability in the data, model complexity, and incomplete information about future events. Metrics that may help stakeholders understand possible outcomes and confidence associated with each will include confidence intervals, prediction intervals, or probability distributions. I will use visualisations to enhance communication of forecasts and level of uncertainty.

To understand key information required for ventilatory decision-making and design visualisations, I will undertake co-design[82] workshops with clinicians who work on intensive care and use the ventilator. These workshops will follow published guidance and involving patients and clinicians throughout[83,84], and will be led by an experienced facilitator, collaborator Dr Tim Stephens. The iterative process of development, testing, modification and refinement will utilise findings from Work Package 1 to inform the design of the early prototype with potential users. We will run three workshops, using a series of co-design techniques to facilitate discussion and elicit design input from clinicians and patients. These will include storyboard techniques (e.g. use of scenarios to illustrate how the tool might work in practice) to test feasibility, display of visualisations for ventilation data to test acceptability, and “think-aloud” techniques, where users are asked to talk about what they are doing while interacting with the visualisation to understand decision-making[85].

I will recruit 20 clinicians from five intensive care units of varying specialisation, including a respiratory failure centre (Royal Brompton Hospital), a paediatric intensive care unit (St Mary’s Hospital), and general

adult intensive care units (Royal London Hospital, Hammersmith Hospital, Kings College Hospital). Agreements are in place with all included units. I will purposively sample to ensure a mix of sex, ethnicity, professional background and clinical experience. Sampling from different units will allow an assessment of unit-based practice, and how visualisations might be used in different settings. While we acknowledge consensus may never be achieved, we believe this number of clinicians represents an adequate subset to sample based on prior work by our group. I will include our PPIE co-applicant, as well as 2 representatives from our Patient Family Advisory Group, in the workshops. We anticipate each workshop to last half a day, and each will include approximately 7-8 participants. Our experienced facilitator will ensure equal participation and a flattened hierarchy.

Co-design workshops will be designed with the assistance of the Imperial Healthtech Research Centre and collaborator Dr Stephens, overseen by supervisor Dr Ni. Activities and questions will be piloted before use. Workshops will be video recorded, and field notes will be taken contemporaneously. The discussions will be transcribed, coded and undergo thematic analysis based on Braun and Clarke's reflexive thematic analysis[86] using a hybrid of inductive and deductive approaches. Data analysis will be conducted by myself, Dr Stephens and Dr Ni, with identified codes and themes compared for concordance or disagreement.

Feedback from the first co-design workshop will be used to design visualisations (e.g., time series plots, scenario analysis charts, heat maps, temporal attention plots, interactive dashboards) to display different aspects of the model considered important for decision-making, supervised by Dr Serban. Visualisations will be showcased through the Data Observatory at

the DSI, where a second round of co-design will be undertaken with the same groups to showcase initial design and gather feedback, assessing acceptability of the designs. The second round of co-design workshops will be designed, performed and analysed in a similar fashion to round 1.

Following this, I will create a user interface (UI), using a system architecture that incorporates and processes the data, and a web front-end to display the results, built around free-lock pipelines with reusable components that transform data into a desirable format (allowing each component to be efficient)[27,30]. Tasks will be run in the background due to required data and processing volume. Processed data will output into a front-end data store (PostgreSQL, an open-source relational database management system); the front-end UI will run as HTML5 and React.js JavaScript app. The UI will enable interaction with and filtering of data, and will be presented to the co-design groups using validated tools[87], testing feasibility and optimising for use in a clinical setting.

This approach will seek to establish a deeper understanding of acceptability of the intervention, usability needs for the interface, and implementation requirements. This collaboration will involve our NHS partners; Dr Serban and the DSI to assist with visualisation design; Dr Ni the NIHR Healthtech Research Centre (HRC) to assist with qualitative study design, analysis, and UI development; Dr Tim Stephens as a collaborator with expertise in co-design; and clinical engineering at Aalborg University to help with model and UI design.

Objective 3:

- To assess how visualisations alter clinical decision-making in a simulated environment.
- To assess how visualisations affect decision confidence in a simulated environment.

To assess impact, I will conduct a human-AI interaction vignette study for clinicians that operate the ventilator using a modified, between-subjects design. Those recruited will be required to have at least 5 years of clinical experience on intensive care and regularly adjust the ventilator in clinical practice. We will purposively recruit to ensure a mix of professional background and experience. A power calculation has shown that we will require x clinicians to demonstrate…

Each subject in the experiment will perform 16 trials, with the first four being used as a training period and thus identical for each participant. The subsequent 12 trials will comprise the main experiment; for each trial, subjects will be asked to determine whether they would wean the patient from the ventilator, and to decide on ventilator mode and settings for the next hour. Trials will be selected from real-world patient data from the analyses in Work Package 1, by Dr Patel and Dr Ramnarayan, mechanical ventilation experts, and will represent all four ventilation time stratifications, thus ensuring a representative mix of complexity.

We will use a multifactorial experimental design with three arms, where each clinician will experience all three arms on different subsets of patients. The three arms are: baseline with no additional AI information (baseline); additional AI decision support system information (AI);

additional AI decision support system with explanation of AI decision using data visualisation (XAI). Our primary measure of interest will be difference in ventilator settings for the same patient between the 3 groups, as a measure of the impact that the arm has on clinical decisions. Clinicians will also be asked to engage in think-aloud techniques, expressing rationale for their decisions and perceived confidence in decision-making. Simulations will be audio recorded. I will employ statistical analyses including two-sided T-tests for comparison of means, and linear regression for assessing associations, and audio recordings will be transcribed, coded and undergo thematic analysis based on Braun and Clarke's reflexive thematic analysis[86].

6. Dissemination, outputs and anticipated impact

On completion of each objective, we will disseminate our findings through local scientific meetings, national and international conferences (e.g., the American Thoracic Society Annual Congress and European Society for Intensive Care Medicine Annual Congress), and through publication. As part of a research group which has published in high impact journals, and with supervisors from diverse disciplines, I will be supported publish in peer-reviewed general medical, health services and data science journals. I have an academic track record of 12 publications, several conference abstracts, and have co-edited and written several chapters for a medical textbook, published by Cambridge University Press; additionally, my year working as the *Anaesthesia* Trainee Editor Fellow has given me strong insight into what is sought by editorial teams reviewing papers submitted to anaesthesia and critical care journals.

I will utilise established NIHR Healthtech Research Centre and Imperial College networks of industry, clinician, research, public and patients, and other key stakeholders to identify areas for shared learning (e.g., Severe Acute Respiratory Failure Network and Imperial AI Network). I have established collaborations with Aalborg University, the VentilaGroup study team in Spain, and with UK groups, including two specialist severe respiratory failure and long-term weaning centres. We have received assistance from the Intensive Care Society, the Paediatric Critical Care Society and Critical Care Research Group and will continue to use these networks to disseminate our work. Our research group at Imperial also has a history of collaboration with several international groups. Work will be shared the aim of receiving feedback and sharing methodology; I will ensure reproducibility of our findings by sharing code through repositories such as Zenodo and in the supplementary materials of accepted publications.

We will liaise with our Patient and Family Advisory Group to determine the best methods for dissemination to the public, while asking them to review and contribute to any written material or media, and to co-present our findings at conferences. We will utilise the Imperial College Science Media Team to publicise our work, as well as disseminating through groups such as COVID support groups and the British Lung Foundation. We will also make use of public outreach activities through Imperial College London.

I anticipate this research will have significant impact for patients, clinicians, and the NHS. Patients will benefit through receipt of efficacious, evidence-based therapy at the right time, improving safety and reducing ventilator complications. Clinicians will benefit through improved understanding of ventilatory weaning, thereby reducing heterogeneity in practice, and

improving clinical training, shared learning, and communication with patients and families. Finally, the NHS will benefit through optimisation of patient time on the ventilator, reducing length of stay and cost of care.

7. Project management

I will be working full-time on this research project, under the supervision of Dr Brijesh Patel, Clinical Senior Lecturer in cardiothoracic critical care within the Anaesthesia, Pain Medicine and Intensive Care (APMIC) section of the Department of Surgery and Cancer, Faculty of Medicine at Imperial College London, with whom I will have weekly meetings to monitor progress, and who will maintain high-level oversight and guidance.

I will be co-supervised by Dr Padmanabhan Ramnarayan, Clinical Reader in Paediatric Critical Care within APMIC at Imperial, who will provide guidance on paediatric ventilation and lend his experience of clinical informatics and artificial intelligence; Dr Melody Ni, decision analyst and methodologist and Deputy Director of the NIHR Healthtech Research Centre (HRC) at Imperial, who will be responsible for offering supervision on the qualitative, methodological and implementation aspects, has expertise in developing methods to translate technology into clinical practice, and provides links to NIHR centres, including HRCs; and Dr Ovidiu Serban, Research Fellow at the Imperial Data Science Institute (DSI), who has expertise in machine learning and largescale data visualisation and will provide a link to the Data Observatory at the DSI.

Additionally, I will be supported by collaborators including Professor Steve Rees, Professor of Respiratory Critical Care Technology at Aalborg University, who brings expertise in clinical decision support systems and biomedical engineering; Dr Tim Stephens, Clinical Lecture and

Implementation Scientist at Queen Mary University of London, who brings expertise in qualitative research and methodology; and NHS partners, including Dr TC Aw, Associate Chief Clinical Information Officer for Clinical Analytics & Informatics at the Royal Brompton Hospital, an expert in real-world clinical data development leading to application opportunities for research, digital technology development, decision-making support algorithms and training datasets for AI, and Professor Luigi Camporota at Guy's & St Thomas' NHS Trust, who has extensive clinical and academic expertise in mechanical ventilation.

8. Ethics

Several broad issues relating to artificial intelligence (AI) in healthcare apply to this project. These include patient confidentiality, data security, data transparency and accountability. In brief, this project is a retrospective observational study. Data used for analysis has either been anonymised and stored on a Google Cloud platform (MIMIC-IV), which meets required data protection security standards; is pseudo-anonymised, stored on secure servers with access obtained by following appropriate organisational procedures, and will be anonymised prior to use for analysis (GSTT, RBHT and ICHT databases); or has already undergone review and obtained ethical approval for use in ventilation analyses (aggregated paediatric database, IRAS 323325). I will apply to the Health Research Authority for approval for use of NHS data, but following assessment with the online HRA tool, Research Ethics Committee review is likely unnecessary. Patient consent will not be required as all data is or will be anonymised. Data sharing agreements are in place for all databases, which have undergone appropriate processing for quality standards. Machine learning analysis trains data on the whole patient cohort, so risk of identification is further

reduced. Data at Imperial is stored on a cloud-based system that is authenticated to meet the College's standards for data security and is compliant with UK and EU Data Protection Laws. To minimise risk of corruption or unauthorised access, Jupyter notebooks will be kept within the Imperial laboratory. All code used to extract and process data will be maintained in Jupyter notebooks, stored in the online data repository, Zenodo, and shared through conference abstracts, manuscript pre-prints, and publications, improving transparency and allowing other researchers to review and replicate our analysis.

Aware of the potential for AI algorithms to perpetuate algorithmic bias, exacerbating inequalities by replicating established social and racial biases, I will review the algorithms for bias and assess for calibration in under-represented groups. I will assess the algorithms against international standards for ethical machine learning and ensure reporting of results conform to SAGER guidelines and the NIHR-INCLUDE Ethnicity Framework. I will purposively sample for the focus groups to ensure a distribution of age, sex, and ethnicity that reflects the population.

Alongside presenting to several public panels and seeking feedback from the Research Partner's Group, Research Design Service and Patient Experience Research Centre at Imperial, and charities ICUSteps and Voice, I have convened a 12-member Patient and Family Advisory Group, with lived experience of invasive mechanical ventilation to advise on study design, data analysis, ethical issues relating to our findings, and dissemination of results. Their feedback has been very positive, with panel members agreeing this is an important topic with potential for substantial impact; their feedback has been instrumental in designing this proposal.

9. Success criteria

Success of this project will ultimately be judged by the dissemination of outputs as detailed in this project plan, and completion of the doctorate. Additionally, the success of each objective will be judged as follows:

1. Iteratively develop and validate XAI-based models to delineate subgroups based on physiological trajectories.

- Identify and validate weaning trajectories using time-series machine learning derived from routinely collected clinical data.
- Identify features of importance in the model and how these vary with time.
- Successfully build a predictive classification model that can identify short-, medium, and long-term trajectories from routine clinical data.
- Demonstrate how clinical interventions (e.g., ventilatory, pharmacological) affect predictions.
- Review model for bias and assess for calibration in under-represented groups.

2. Develop and optimise data visualisation approaches to improve model transparency for clinical application.
    - Define key parameters required for ventilatory weaning and decision-making.
    - Develop visualisations that intuitively display data.
    - Develop a user interface that enables interaction with the data.
    - Optimise for clinical application through stakeholder review.

3. Validate data visualisation approach and investigate factors required for clinical implementation.
    - Assess impact of XAI with visualisation on clinical decision-making when compared with normal practice and XAI suggestion without visualisation.
    - Assess impact of XAI with visualisation on decision-making confidence.

10. Risk mitigation

One of the major problems with using retrospective observational data relates to data quality. MIMIC-IV is a high granularity database that is accepted as a reliable data source in the literature, with good documentation and harmonisation between datasets, and over 2,000 publications to date. All code used for generating tables for analysis has been made publicly available through Github (https://github.com/MIT-LCP). The ICHT, RBHT, GSTT and paediatric critical care databases have undergone appropriate processing for quality standards and where further data is required these methods will be followed; database curation for the NHS databases has been costed into this PhD application.

The application and testing of three neural network architectures (Temporal Fusion Transformer, Temporal Convolutional Networks and Long-Short Term Memory Nets) to identify weaning trajectories will mitigate against the risk of model failure or poor performance.

The involvement of supervisors and collaborators from several disciplines should mitigate against failure to complete the research objectives.

7. Patient and public involvement

Please describe how patients/ services users, carers and the public have been involved in developing this proposal. (350 words)

I have engaged members of the public through the charities ICUSteps (https://icusteps.org/) and VOICE (https://www.voice-global.org), and through Imperial College London, presenting at the Research Partner's Group (RPG) and the Patient Experience Research Centre (PERC), to those with lived experience of adult and paediatric ventilation. I have received feedback from the Research Support Service (RSS), who consult members of the public. Finally, recruiting through VOICE and PERC, I have convened a 12-person Patient and Family Advisory Group (PFAG), and a co-applicant, Ms Joanne Lloyd, who will advise on the project.

Presentations to the RPG and PERC has involved 3 iterations of the project; I presented to 6 panel members at the RPG, and 2 panel members through PERC, who provided engaging feedback. All agreed this was a very important research topic which could significantly benefit for patients and the NHS. This was reflected in feedback from the RSS, which highlighted the potential to reduce variation in clinical practice.

The lay summary initially contained jargon and objectives were unclear; after the input of the RPG, PERC and the PFAG, it was considered to have captured the project's main themes and was easy to understand. The lay summary was also sent to members of the charity ICUSteps, and feedback was very positive, with one respondent commenting, "It's clear ***what*** the research wants to look at, why they are doing it and how they

will do it. It is particularly encouraging to see the development of the need to understand the patient as an individual."

Following presentations to the RPG and PERC, themes to emerge from the discussion were:

- Potential for artificial intelligence to perpetuate bias.
- Data security and maintaining confidentiality.
- Importance of transparency when reporting results.
- Ventilation outcomes that are important to patients.
- Potential for visualisations to communicate complex information to patients and family members.
- Guidance on recruiting lay panel:
    - Importance of including panel members with lived experience of intensive care.
    - Ensuring diversity amongst panel members.
    - Ensuring appropriate safeguards in place.

This has led to amendment of this proposal, and following discussion, we allayed these concerns and recruited the PFAG.

Please describe the ways in which patients/ service users, carers and the public will be actively involved in the proposed research, including any training and support provided. (350 words)

I have convened a PFAG to advise on the project. Feedback from the RPG was instrumental for reducing barriers to participation; where I had initially considered good written and spoken communication skills, and

recruiting on a first come, first served basis, the RPG suggested this could led to a homogenous group and exclude certain voices. Subsequently, criteria for recruitment were that participants should have lived experience of intensive care, either as a patient or family member, should be in stable recovery of at least 3 months since ICU contact; and > 18 years old. The group we have purposively convened consists of 12 people, age range 34-74, 75% female, 25% non-white, from all over the UK, and with lived experience of adult and paediatric ventilation.

Their feedback has been instrumental in developing the proposal. We will meet at least four-monthly over video conference to discuss our findings and guide the project. The PFAG will also assist with crafting our message for public dissemination and contribute to written material. We will support PFAG members with development opportunities through PERC. One member of the group, Ms Joanne Lloyd, who has experience as a PPIE member for other mechanical ventilation studies, has agreed to join the study as a co-applicant. Her expertise will be invaluable in assisting with co-design workshops, co-presenting results at conferences, and ensuring the patient voice remains at the forefront.

Additionally, we will use the Imperial College Science Media Team to publicise our work, and disseminate through groups such as the British Lung Foundation and ICUSteps to reach a wider audience.

For PPIE training and support, I will seek support from PERC, including attending their 4-week Public Involvement Training course, and seek support from my supervisors and collaborators, all of whom have experience of involving patients in research. This proposal has also

costed the time of the PFAG and co-applicant in accordance with the NIHR Centre for Engagement and Dissemination guidelines.

To evaluate PPIE involvement and capture its impact, I will maintain all patient involvement in the PERC log and regularly evaluate progress with the PFAG and my supervisors.

If it is considered not appropriate and meaningful to actively involve patients/ service users, carers and the public in your proposed research, please justify why. (350 words)

N/A

8. Training and development and research support

1. **Proposed training and development programme (1000 words).**

   I am an early career researcher, highly motivated to become an expert in the field of invasive mechanical ventilation (IMV), using mixed methodology to develop clinical decision support tools using explainable artificial intelligence (XAI). I aim to use the opportunity of this prestigious NIHR Doctoral Fellowship as a springboard to launch a career as a Clinician Scientist, lead my own research group and attain a Chair in anaesthesia and critical care.

   Supervision

   In the short-term, I intend to complete a high-quality PhD under the guidance of Dr Brijesh Patel and Dr Padmanabhan Ramnarayan, physician scientists in adult and paediatric critical care and ventilatory support; Dr Melody Ni, decision analyst and methodologist with a background in translating innovation into clinical practice; and Dr Ovidiu Serban, senior data scientist with expertise in machine learning and largescale visualisation projects. The expertise in my supervisory team will empower me to deliver on this ambitious project while acquiring new research skills, disseminate my work through high-impact publications and develop into an independent researcher. Gaining Fellowship support from the NIHR will allow me to focus on these goals. Embedding within the Imperial Data Science Institute (DSI), I will learn how to apply novel time-series machine learning to understand weaning trajectories and use novel data visualisations to deliver on the goal of XAI. I will obtain support from the Imperial-NIHR Healthteach Research Centre (HRC) to learn, develop and assess optimal approaches to improve clinical implementation. This is a novel approach for producing XAI tools. I will

draw on my background in data science, the experience of my supervisory team, and collaborators: machine learning (Dr Parbhoo at Imperial), ventilation (Dr Camporota at Kings College Hospital), qualitative research (Dr Stephens at Barts Health), and decision support development (Professor Rees at Aalborg University), to deliver on these goals.

Courses

I will attend local courses at Imperial College, accessed through the Clinical Academic Training Office, who provide academic and pastoral support for clinical PhD students, and host courses ranging from statistical analysis to academic writing. I will attend the Public Involvement Training course, run through the Patient Experience Research Centre at the start of my Fellowship.

To develop skills in qualitative methodology, implementation science and co-design, I have costed the “Qualitative Research Methods in Health” course at University College London, “An Introduction to Implementation Science” at Kings College London, and “Experience-Based Co-design”, delivered by the Point of Care Foundation. Additionally, following supervisory recommendation, I will attend the “Developing & Evaluating Complex Interventions” course, delivered online by the University of Glasgow. This will be important for understanding how complex interventions (clinical decision support for IMV) can be implemented in a complex environment (critical care).

To improve skills in machine learning and coding, I have costed annual membership for Coursera (www.coursera.org) and Datacamp

(www.datacamp.com). For example, on Coursera, the “AI for Medicine” course includes modules on time to event modelling, useful for planned analyses. Courses on Datacamp will be key for improving coding skills. I will attend the “Mathematics for Machine Learning” from the MSc in Computing at Imperial, which is included as part of Coursera membership. Imperial College runs courses as part of the UKRI AI for Healthcare Centre for Doctoral Training. I will utilise these to address gaps in my knowledge.

As a basis for using visualisations to improve understanding of complex data, I will attend the “Data Visualisation and Visual Analytics” course at the Alan Turing Institute, which is delivered online.

International collaboration

I will visit Aalborg University to mature a collaboration with Professor Steve Rees and his team. Professor Rees is an expert in biomedical engineering, decision support systems and physiological modelling. During an introductory visit in August 2022, we worked on a preliminary analysis using physiological data to subgroup patients weaning from the ventilator. We have subsequently worked on a narrative review assessing clinical practice and decision-making in patients requiring IMV (published in the *British Journal of Anaesthesia*).

This visit, costed into the PhD, will be key for developing skills in data analysis, machine learning, and decision support, enriching my development and that of our critical care research group in London, while fostering ongoing academic relationships.

The future

On completion of the PhD, I will apply for a post-doctoral fellowship (e.g., NIHR Clinical Lectureship) to develop as an independent researcher with a formal arrangement for postdoctoral training. Clinically, I continue Anaesthesia and Intensive Care Medicine registrar training, where I will work towards my Certificate of Completion of Training as a dual specialist. I will continue to seek guidance from supervisors and mentors.

Long-term, I am motivated to follow a clinical academic track. I intend to obtain a clinical academic faculty position in a university and associated hospital, facilitating my research goals. At present, my academic interests are focused on using machine learning to improve understanding of ventilation and develop tools that improve patient outcomes, while collaborating with laboratory scientists to understand the biological basis for our findings. The field of data science in critical care is expanding but limited work has been performed in ventilation. This Fellowship will enable the curation of one of the largest longitudinal datasets for critical care globally, enabling future analyses and creating an open dataset for other investigators and collaborators.

Securing this NIHR Fellowship will provide the necessary protected time to immerse myself in this research. The Fellowship will offer accelerated personal and career development to develop research skills, while providing access to the NIHR infrastructure, such as career development schemes, mentorship, and networking opportunities, aiding development as an independent investigator. The Fellowship will allow me to work within a world-leading critical care research group at Imperial, while accessing collaboration opportunities. The support and

endorsement of the NIHR will be invaluable during dissemination of our findings to the international research community, clinicians, patients, and the public, which will enhance my clinical, data science, and academic networks. All these experiences will strengthen my academic portfolio and help with the transition to working as an independent researcher.

2. **Research Support/ Supervisor (250 words).**

   Dr Brijesh Patel

   Dr Patel was my supervisor during my NIHR Academic Clinical Fellowship, and will be my primary supervisor during the PhD, maintaining an overview of the project. He has a background in basic and translational research in respiratory failure and mechanical ventilation, with an interest in understanding the biological and physiological mechanisms of failure to wean from the ventilator. He has been awarded > £5 million in translational critical care grants, and is recognised as an international expert in the field, with > 80 publications and several national and international awards. He led the national service evaluation during the COVID-19 pandemic, which used explainable artificial intelligence to improve understanding of this disease. He has supervised 7 PhD candidates, of which 4 have been awarded a PhD, 1 in write up phase, and he is currently supervising 2 candidates. He has also led two clinical trials (in collaboration with Professor Rees), on the implementation of clinical decision support systems in critical care, which have led to the development of this proposal. Furthermore, he is principal investigator for a NIHR EME funded RCT examining mechanical ventilation strategies in patients with severe respiratory failure supported on extracorporeal life support, and studies examining ICU-acquired muscle wasting, a critical determinant and consequence of failure to wean. I will meet with

Dr Patel on a weekly basis, and he will provide support on all aspects of this project, maintaining high-level oversight.

Dr Padmanabhan Ramnarayan

Dr Ramnarayan's research interests are focused on the use of artificial intelligence to improve decision-making in critically ill children, use of novel non-invasive imaging techniques such as lung ultrasound during ventilation, and pragmatic clinical trials, particularly related to paediatric ventilation. He led a master protocol of two national clinical trials on non-invasive ventilation in children (FIRST-ABC) and he is currently leading a clinical trial of respiratory support in acute bronchiolitis (BACHb). He has an active interest in clinical informatics, which started during his MD(Res) research conducting a series of studies examining the accuracy and utility of a computerised diagnostic decision support system. He has been awarded £4 million in critical care research grants as Chief Investigator and ~£14 million as Co-Investigator, and he is recognised as an international expert in the field, with over 150 publications. He is currently supervising 4 PhD candidates and 1 MD(Res) candidate, one of whom is due for completion in 2024. Dr Ramnarayan will provide support on the paediatric ventilation aspects of this project, which has traditionally been considered separately to adult ventilation, as well as lending his expertise on decision support systems.

Dr Melody Ni

Dr Ni is a decision analyst and a methodologist with over 17 years of experiences in generating multi-dimensional evidence for the translation of innovative ideas into clinical practice. Dr Ni is a deputy director of the NIHR HRC in IVD at Imperial and has been leading a team of multi-

disciplinary researchers since 2013, delivering impact through a range of innovative projects, including the development of a breath test for the detection of gastrointestinal cancers. Dr Ni chairs a regulatory network for in vitro diagnostics, working with industry partners and key stakeholders including MHRA and NICE. She has also hosted multiple accelerator programmes aimed at building knowledge and expertise for clinical scientists and innovators. Dr Ni has won over £10 million in grant funding, recognised as an international expert in the field, with over 60 peer-reviewed publications. She has supervised 7 PhD students, including 2 as lead supervisor, of which 6 have been awarded a PhD and 1 an MD. She is currently supervising 3 PhD candidates, 1 as lead supervisor. Dr Ni will provide expertise and guidance on the methods and approach of the project, both quantitatively and qualitatively, focusing on the real world aspect of the technology and the value a solution can realistically deliver. She also provides links to NIHR centres including the HRCs.

Dr Ovidiu Serban

Dr Serban is a Research Fellow at the Imperial Data Science Institute and the Data Observatory group lead. His research interests include machine learning, natural language processing, affective computing and interactive system design. His work includes real-time natural language processing, data curation and large-scale visualisation systems. He holds a joint PhD from INSA de Rouen Normandy (France) and "Babes-Bolyai" University in Romania while working at the LITIS Laboratory in France. He has over 40 peer-reviewed publications, including one best paper award at a NeurIPS workshop. He has supervised 3 PhD students as lead supervisor, of which two are close to their viva, and one as a co-supervisor. He supervised over 40 MSc students at Imperial College, with

over 70% receiving distinction or distinguished, including three students being awarded president or department awards. Dr Serban will oversee the project's machine learning and data visualisation aspects and provide a link to the Data Observatory at the DSI.

3. **Collaborations (600 words)**

I have built several important collaborations. I will partner with the Imperial Data Science Institute (DSI) on the data science and visualisation aspects. Their aim is to advance research and create a community in the field of data science; the unifying theme of their research is the pursuit of innovations for building, exploring, and exploiting modern datasets. They host researchers from multiple disciplines, including Biomedical Informatics, Data Visualisation and AI System Architecture. Specifically, I will work with Dr Ovidiu Serban, an expert in machine learning and large-scale visualisation, and Dr Kai Sun, who has experience managing translational research projects at the DSI. They will provide in-kind access to the Data Observatory throughout the project for visualisation testing. This collaboration supports two of the core objectives of the DSI (developing data analysis technology that supports research and translating data science into innovation through public sector partnership); it also supports the DSI's portfolio expansion in the field of healthcare data, while leading to new relationships with academics from Imperial College London.

I will collaborate with the NIHR Healthtech Research Centre (HRC), whose aim is to support the development of medical devices, diagnostics, and digital technologies, which help clinicians and patients to monitor, diagnose and manage conditions including cancer and cardiovascular

and respiratory diseases. Their multidisciplinary team includes experts in clinical, technology, human factors and health economics, who drive innovation, advance new technology and support healthcare staff to reduce workload. I will be supervised by Dr Melody Ni, deputy director of the NIHR-HRC. The HRC will also assist with building networks with world-leading hospitals and clinical academics. My project will add to their portfolio of innovation using artificial intelligence and the development of decision-aiding medical devices.

I will mature an exciting collaboration with Aalborg University, which was established during an introductory visit in August 2022. I worked with Professor Steve Rees, Professor of Respiratory Critical Care Technology in the Department of Health Science and Technology, on a preliminary analysis using physiological data to subgroup patients weaning from the ventilator in the Weaning Algorithms for mechanical VEntilation (WAVE, NCT 03697785) study. Subsequently, we have worked on a narrative review assessing clinical practice and decision-making in patients on the ventilator (published in the *British Journal of Anaesthesia*). Professor Rees and his research team have led the development of a model-based clinical decision support system for mechanical ventilation, the Beacon Caresystem (doi: 10.1109/EMBC.2015.7319595), which has been shown to provide appropriate ventilator settings, reduce pressure support whilst preserving respiratory function, and been the subject of several international randomised controlled trials, during which they have collaborated with clinicians at Imperial College London, including my primary supervisor, Dr Brijesh Patel. I intend to mature this collaboration during the PhD with a second visit to Aalborg, which has been costed into this proposal, to develop expertise in data analysis techniques and decision support; the Aalborg team provide expertise in biomedical

engineering, physiological modelling, and decision support systems. Aalborg University foresee several advantages to the collaboration, including improved understanding of respiratory physiology, exploring novel methods for improving explainability of decision support systems, while fostering ongoing international networks.

Finally, I have several partnerships with NHS and university collaborators, built during my Academic Clinical Fellowship, including with Dr TC Aw at the Royal Brompton Hospital, an expert in real-world clinical data development, Professor Luigi Camporota, who has extensive clinical and academic expertise in mechanical ventilation, and Dr Tim Stephens, Clinical Lecture and Implementation Scientist at Queen Mary University of London, who brings expertise in qualitative research and methodology. Their guidance and support will be key to ensuring the success of this Fellowship.

9. Detailed budget

See application

References

1. Fan E, Brodie D, Slutsky AS. Acute Respiratory Distress Syndrome: Advances in Diagnosis and Treatment. *JAMA* 2018; 319: 698.

2. ICNARC. *Key statistics from the Case Mix Porgramme - adult, general critical care units*. Intensive Care National Audit & Research Centre, https://www.icnarc.org/Our-Audit/Audits/Cmp/Reports/Summary-Statistics (5 May 2022, accessed 19 November 2022).

3. PICANet. *PICANet State of the Nation Report 2023*, https://www.picanet.org.uk/annual-reporting-and-publications/ (2023).

4. Ventilation with Lower Tidal Volumes as Compared with Traditional Tidal Volumes for Acute Lung Injury and the Acute Respiratory Distress Syndrome. *N Engl J Med* 2000; 342: 1301–1308.

5. Guérin C, Reignier J, Richard J-C, et al. Prone Positioning in Severe Acute Respiratory Distress Syndrome. *N Engl J Med* 2013; 368: 2159–2168.

6. The RECOVERY Collaborative Group. Dexamethasone in Hospitalized Patients with Covid-19. *N Engl J Med* 2021; 384: 693–704.

7. on behalf of the United Kingdom COVID-ICU National Service Evaluation, Patel BV, Haar S, et al. Natural history, trajectory, and management of mechanically ventilated COVID-19 patients in the United Kingdom. *Intensive Care Med* 2021; 47: 549–565.

8. Morris AH, Stagg B, Lanspa M, et al. Enabling a learning healthcare system with automated computer protocols that produce replicable and personalized clinician actions. *J Am Med Inform Assoc* 2021; 28: 1330–1344.

9. Karbing DS, Spadaro S, Dey N, et al. An Open-Loop, Physiologic Model–Based Decision Support System Can Provide Appropriate Ventilator Settings: *Crit Care Med* 2018; 46: e642–e648.

10. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *Npj Digit Med* 2020; 3: 17.

11. Clavieras N, Wysocki M, Coisel Y, et al. Prospective Randomized Crossover Study of a New Closed-loop Control System *versus* Pressure Support during Weaning from Mechanical Ventilation. *Anesthesiology* 2013; 119: 631–641.

12. Lellouche F, Mancebo J, Jolliet P, et al. A Multicenter Randomized Trial of Computer-driven Protocolized Weaning from Mechanical Ventilation. *Am J Respir Crit Care Med* 2006; 174: 894–900.

13. East TD, Heermann LK, Bradshaw RL, et al. Efficacy of computerized decision support for mechanical ventilation: results of a prospective multi-center randomized trial. *Proc AMIA Symp* 1999; 251–255.

14. Su J, Lin C-Y, Chen P-J, et al. Experience with a step-down respiratory care center at a tertiary referral medical center in Taiwan. *J Crit Care* 2006; 21: 156–161.

15. Hsu J-C, Chen Y-F, Chung W-S, et al. Clinical Verification of A Clinical Decision Support System for Ventilator Weaning. *Biomed Eng OnLine* 2013; 12: S4.

16. Zheng K, Kumar S, Sarti AJ, et al. Economic feasibility of a novel tool to assist extubation decision-making: an early health economic modeling. *Int J Technol Assess Health Care* 2022; 38: e66.

17. Capdevila X, Perrigault P-F, Ramonatxo M, et al. Changes in breathing pattern and respiratory muscle performance parameters during difficult weaning: *Crit Care Med* 1998; 26: 79–87.

18. Vassilakopoulos T, Zakynthinos S, Roussos C. The Tension–Time Index and the Frequency/ Tidal Volume Ratio Are the Major Pathophysiologic Determinants of Weaning Failure and Success. *Am J Respir Crit Care Med* 1998; 158: 378–385.

19. Patel B, Mumby S, Johnson N, et al. *Decision support system to evaluate VENTilation in the Acute Respiratory Distress Syndrome*. Preprint, Intensive Care and Critical Care Medicine. Epub ahead of print 29 September 2023. DOI: 10.1101/2023.09.28.23295668.

20. Morris AH. Developing and Implementing Computerized Protocols for Standardization of Clinical Decisions. *Ann Intern Med* 2000; 132: 373.

21. Knighton AJ, Kean J, Wolfe D, et al. Multi-factorial barriers and facilitators to high adherence to lung-protective ventilation using a computerized protocol: a mixed methods study. *Implement Sci Commun* 2020; 1: 67.

22. Sarti AJ, Zheng K, Herry CL, et al. Feasibility of implementing *Extubation Advisor* , a clinical decision support tool to improve extubation decision-making in the ICU: a mixed-methods observational study. *BMJ Open* 2021; 11: e045674.

23. Liao K-M, Ko S-C, Liu C-F, et al. Development of an Interactive AI System for the Optimal Timing Prediction of Successful Weaning from Mechanical Ventilation for Patients in Respiratory Care Centers. *Diagnostics* 2022; 12: 975.

24. Lin M-Y, Li C-C, Lin P-H, et al. Explainable Machine Learning to Predict Successful Weaning Among Patients Requiring Prolonged Mechanical Ventilation: A Retrospective Cohort Study in Central Taiwan. *Front Med* 2021; 8: 663739.

25. Otaguro T, Tanaka H, Igarashi Y, et al. Machine Learning for Prediction of Successful Extubation of Mechanical Ventilated Patients in an Intensive Care Unit: A Retrospective Observational Study. *J Nippon Med Sch* 2021; 88: 408–417.

26. Keim DA. Visual exploration of large data sets. *Commun ACM* 2001; 44: 38–44.

27. Fernando S, Amador Díaz López J, Şerban O, et al. Towards a large-scale twitter observatory for political events. *Future Gener Comput Syst* 2020; 110: 976–983.

28. McGinn D, Birch D, Akroyd D, et al. Visualizing Dynamic Bitcoin Transaction Patterns. *Big Data* 2016; 4: 109–119.

29. Molina-Solana M, Birch D, Guo Y. Improving data exploration in graphs with fuzzy logic and large-scale visualisation. *Appl Soft Comput* 2017; 53: 227–235.

30. Șerban O, Thapen N, Maginnis B, et al. Real-time processing of social media with SENTINEL: A syndromic surveillance system incorporating deep learning for health classification. *Inf Process Manag* 2019; 56: 1166–1184.

31. Islam H, Li X, Zaman Rana MdK, et al. An In-depth Needs Analysis to Design a Data Visualization Dashboard Prototype for Critical Care Setting. In: *2022 Workshop on Visual Analytics in Healthcare (VAHC)*. Washington, DC, USA: IEEE, pp. 01–06.

32. Brich N, Schulz C, Peter J, et al. Visual Analytics of Multivariate Intensive Care Time Series Data. *Comput Graph Forum* 2022; 41: 273–286.

33. Thomas MM, Kannampallil T, Abraham J, et al. Echo: A large display interactive visualization of ICU data for effective care handoffs. In: *2017 IEEE Workshop on Visual Analytics in Healthcare (VAHC)*. Phoenix, AZ: IEEE, pp. 47–54.

34. Key statistics from the Case Mix Programme - adult, general critical care units. 1st January 2019-31st December 2022., https://www.icnarc.org/Our-Audit/Audits/Cmp/Reports/Summary-Statistics (2022, accessed 7 July 2023).

35. Goligher E, Ferguson ND. Mechanical ventilation: epidemiological insights into current practices: *Curr Opin Crit Care* 2009; 15: 44–51.

36. Wunsch H, Linde-Zwirble WT, Angus DC, et al. The epidemiology of mechanical ventilation use in the United States*: *Crit Care Med* 2010; 38: 1947–1953.

37. Gunasekera P, Gratrix A. Ventilator-associated pneumonia. *BJA Educ* 2016; 16: 198–202.

38. Jubran A, Grant BJB, Duffner LA, et al. Long-Term Outcome after Prolonged Mechanical Ventilation. A Long-Term Acute-Care Hospital Study. *Am J Respir Crit Care Med* 2019; 199: 1508–1516.

39. NHS England. NHS Long Term Workforce Plan, https://www.england.nhs.uk/publication/nhs-long-term-workforce-plan/ (2023, accessed 3 December 2023).

40. The NHS Long Term Plan, https://www.longtermplan.nhs.uk/ (2019, accessed 7 July 2023).

41. Health Education England. The Topol Review, https://topol.hee.nhs.uk/ (2019, accessed 3 December 2023).

42. Tume LN, Kneyber MCJ, Blackwood B, et al. Mechanical Ventilation, Weaning Practices, and Decision Making in European PICUs*: *Pediatr Crit Care Med* 2017; 18: e182–e188.

43. Blackwood B, Junk C, Lyons JDM, et al. Role Responsibilities in Mechanical Ventilation and Weaning in Pediatric Intensive Care Units: A National Survey. *Am J Crit Care* 2013; 22: 189–197.

44. Rose L, Blackwood B, Egerod I, et al. Decisional responsibility for mechanical ventilation and weaning: an international survey. *Crit Care* 2011; 15: R295.

45. Alkhathami M, Al Haddad M, Alenazi M. PERCEIVED RESPONSIBILITY FOR MECHANICAL VENTILATION AND WEANING DECISIONS IN INTENSIVE CARE UNITS IN THE KINGDOM OF SAUDI ARABIA. *Chest* 2022; 161: A497.

46. Villa G, Manara D, Palese A. Nurses' near-decision-making process of postoperative patients' cardiosurgical weaning and extubation in an Italian environment. *Intensive Crit Care Nurs* 2012; 28: 41–49.

47. Rose L, Nelson S, Johnston L, et al. Workforce profile, organisation structure and role responsibility for ventilation and weaning practices in Australia and New Zealand intensive care units. *J Clin Nurs* 2008; 17: 1035–1043.

48. Curtis B, Rak K, Richardson A, et al. Understanding Interprofessional Decision-Making Processes and Perceptions of Oxygenation for Acute Respiratory Failure. In: *D106. STAFFING, TRAINING, AND IMPLEMENTATION TO IMPROVE ICU QUALITY*. American Thoracic Society, pp. A7814–A7814.

49. Haugdahl HS, Storli S, Rose L, et al. Perceived decisional responsibility for mechanical ventilation and weaning: a Norwegian survey: A Norwegian survey. *Nurs Crit Care* 2014; 19: 18–25.

50. Taylor F. A comparative study examining the decision-making processes of medical and nursing staff in weaning patients from mechanical ventilation. *Intensive Crit Care Nurs* 2006; 22: 253–263.

51. Balas MC, Tate J, Tan A, et al. Evaluation of the Perceived Barriers and Facilitators to Timely Extubation of Critically Ill Adults: An Interprofessional Survey. *Worldviews Evid Based Nurs* 2021; 18: 201–209.

52. Arnal J-M, Garnero A, Novotni D, et al. Closed loop ventilation mode in Intensive Care Unit: a randomized controlled clinical trial comparing the numbers of manual ventilator setting changes. *Minerva Anestesiol*; 84. Epub ahead of print January 2018. DOI: 10.23736/S0375-9393.17.11963-2.

53. Kydonaki K, Huby G, Tocher J, et al. Understanding nurses' decision-making when managing weaning from mechanical ventilation: a study of novice and experienced critical care nurses in Scotland and Greece. *J Clin Nurs* 2016; 25: 434–444.

54. Tingsvik C, Johansson K, Mårtensson J. Weaning from mechanical ventilation: factors that influence intensive care nurses' decision-making: Intensive care nurses' decision-making when weaning from mechanical ventilation. *Nurs Crit Care* 2015; 20: 16–24.

55. Lavelle C, Dowling M. The factors which influence nurses when weaning patients from mechanical ventilation: Findings from a qualitative study. *Intensive Crit Care Nurs* 2011; 27: 244–252.

56. Khalafi A, Elahi N, Ahmadi F. Continuous care and patients' basic needs during weaning from mechanical ventilation: A qualitative study. *Intensive Crit Care Nurs* 2016; 37: 37–45.

57. Rose L, Dainty KN, Jordan J, et al. Weaning From Mechanical Ventilation: A Scoping Review of Qualitative Studies. *Am J Crit Care* 2014; 23: e54–e70.

58. Hancock HC, Easen PR. The decision-making processes of nurses when extubating patients following cardiac surgery: An ethnographic study. *Int J Nurs Stud* 2006; 43: 693–705.

59. Gelsthorpe T, Crocker C. A study exploring factors which influence the decision to commence nurse-led weaning. *Nurs Crit Care* 2004; 9: 213–221.

60. Cederwall C-J, Plos K, Rose L, et al. Critical care nurses management of prolonged weaning: an interview study: Prolonged weaning. *Nurs Crit Care* 2014; 19: 236–242.

61. Mårtensson IE, Fridlund B. Factors influencing the patient during weaning from mechanical ventilation: a national survey. *Intensive Crit Care Nurs* 2002; 18: 219–229.

62. Pettersson S, Melaniuk-Bose M, Edell-Gustafsson U. Anaesthetists' perceptions of facilitative weaning strategies from mechanical ventilator in the intensive care unit (ICU): A qualitative interview study. *Intensive Crit Care Nurs* 2012; 28: 168–175.

63. Blackwood B, Wilson-Barnett J, Trinder J. Protocolized weaning from mechanical ventilation: ICU physicians' views. *J Adv Nurs* 2004; 48: 26–34.

64. Tulaimat A, Mokhlesi B. Accuracy and Reliability of Extubation Decisions by Intensivists. *Respir Care* 2011; 56: 920–927.

65. Allerød C, Karbing DS, Thorgaard P, et al. Variability of preference toward mechanical ventilator settings: A model-based behavioral analysis. *J Crit Care* 2011; 26: 637.e5-637.e12.

66. Aboodi MS, Chen J-T, Hope AA, et al. Exploring the Clinical Decision-Making Process in Approaches to Fluid Resuscitation and Vasopressor Administration in Septic Shock: A Qualitative Think-Aloud Study. In: *D25. CRITICAL CARE: HARD TIMES - RESUSCITATING MY PATIENT: FLUID, BLOOD AND OTHER STRATEGIES*. American Thoracic Society, pp. A6004–A6004.

67. Hakim R, Watanabe-Tejada L, Sukhal S, et al. Acute respiratory failure in randomized trials of noninvasive respiratory support: A systematic review of

definitions, patient characteristics, and criteria for intubation. *J Crit Care* 2020; 57: 141–147.

68. Moraes F da S, Marengo LL, Moura MDG, et al. ABCDE and ABCDEF care bundles: A systematic review of the implementation process in intensive care units. *Medicine (Baltimore)* 2022; 101: e29499.

69. Nurses of the Central and General ICUs of Shiraz Namazi Hospital, Sedation an Delirium Group Hospital Universitari de Bellvitge, SPACeR group (Surrey Peri-operative, Anaesthesia and Critical Care Collaborative Research Group), et al. ESICM LIVES 2016: part two: Milan, Italy. 1–5 October 2016. *Intensive Care Med Exp* 2016; 4: 30, s40635-016-0099–9.

70. Calfee CS, Delucchi K, Parsons PE, et al. Subphenotypes in acute respiratory distress syndrome: latent class analysis of data from two randomised controlled trials. *Lancet Respir Med* 2014; 2: 611–620.

71. Calfee CS, Delucchi KL, Sinha P, et al. Acute respiratory distress syndrome subphenotypes and differential response to simvastatin: secondary analysis of a randomised controlled trial. *Lancet Respir Med* 2018; 6: 691–698.

72. Famous KR, Delucchi K, Ware LB, et al. Acute Respiratory Distress Syndrome Subphenotypes Respond Differently to Randomized Fluid Management Strategy. *Am J Respir Crit Care Med* 2017; 195: 331–338.

73. Bellani G, Guerra L, Musch G, et al. Lung Regional Metabolic Activity and Gas Volume Changes Induced by Tidal Ventilation in Patients with Acute Lung Injury. *Am J Respir Crit Care Med* 2011; 183: 1193–1199.

74. Gattinoni L, Caironi P, Cressoni M, et al. Lung Recruitment in Patients with the Acute Respiratory Distress Syndrome. *N Engl J Med* 2006; 354: 1775–1786.

75. Writing Group for the Alveolar Recruitment for Acute Respiratory Distress Syndrome Trial (ART) Investigators, Cavalcanti AB, Suzumura ÉA, et al. Effect of Lung Recruitment and Titrated Positive End-Expiratory Pressure (PEEP) vs Low PEEP on Mortality in Patients With Acute Respiratory Distress Syndrome: A Randomized Clinical Trial. *JAMA* 2017; 318: 1335.

76. Briel M, Meade M, Mercat A, et al. Higher vs Lower Positive End-Expiratory Pressure in Patients With Acute Lung Injury and Acute Respiratory Distress Syndrome: Systematic Review and Meta-analysis. *JAMA* 2010; 303: 865.

77. Constantin J-M, Jabaudon M, Lefrant J-Y, et al. Personalised mechanical ventilation tailored to lung morphology versus low positive end-expiratory pressure for patients with acute respiratory distress syndrome in France (the LIVE study): a multicentre, single-blind, randomised controlled trial. *Lancet Respir Med* 2019; 7: 870–880.

78. Bos LDJ, Sjoding M, Sinha P, et al. Longitudinal respiratory subphenotypes in patients with COVID-19-related acute respiratory distress syndrome: results from three observational cohorts. *Lancet Respir Med* 2021; 9: 1377–1386.

79. Toal CM, Fowler AJ, Patel BV, et al. Hypoxemia Trajectory of Non-COVID-19 Acute Respiratory Distress Syndrome Patients. An Observational Study Focusing on Hypoxemia Resolver Status. *Crit Care Explor* 2023; 5: e0985.

80. Tams CG, Ataya A, Euliano NR, et al. Decision support system facilitates rapid decreases in pressure support and appropriate inspiratory muscle workloads in adults with respiratory failure. *J Crit Care* 2017; 42: 213–217.

81. Banner MJ, Euliano NR, MacIntyre NR, et al. Ventilator Advisory System Employing Load and Tolerance Strategy Recommends Appropriate Pressure Support Ventilation Settings. *Chest* 2008; 133: 697–703.

82. Johnson MP, Ballie J, Thorup T, et al. CO/DEsign: building a shared dialogue around analysis within co-design. *Des J* 2017; 20: S4241–S4252.

83. Montori VM, Breslin M, Maleska M, et al. Creating a Conversation: Insights from the Development of a Decision Aid. *PLOS Med* 2007; 4: e233.

84. Coulter A, Stilwell D, Kryworuchko J, et al. A systematic development process for patient decision aids. *BMC Med Inform Decis Mak* 2013; 13 Suppl 2: S2.

85. Wherton J, Sugarhood P, Procter R, et al. Co-production in practice: how people with assisted living needs can help design and evolve technologies and services. *Implement Sci* 2015; 10: 75.

86. Saunders CH, Sierpe A, Plessen C von, et al. Practical thematic analysis: a guide for multidisciplinary health services research teams engaging in qualitative analysis. *BMJ* 2023; 381: e074256.

87. Weiner BJ, Lewis CC, Stanick C, et al. Psychometric assessment of three newly developed implementation outcome measures. *Implement Sci* 2017; 12: 108.